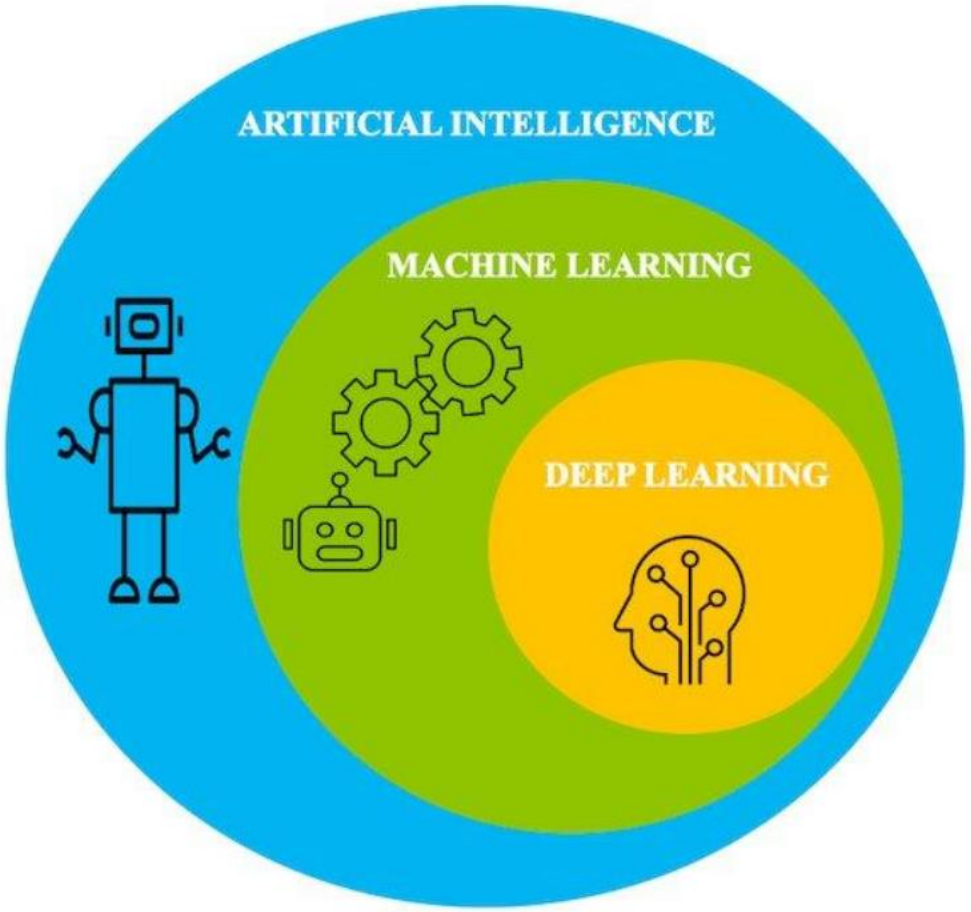
Artificial Intelligence

Lecture: Introduction to Deep Learning



Introduction

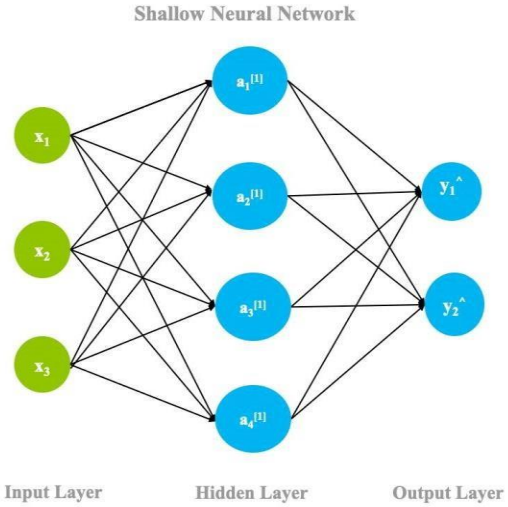
- Deep Learning which is based on artificial neural networks.
- It is capable of learning complex patterns and relationships within data.
- It is based on Artificial Neural Networks (ANN)
- also known as Deep Neural Networks(DNNs)

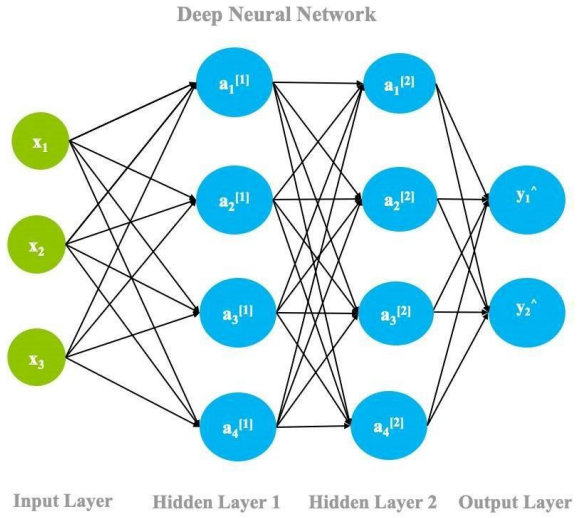
Introduction to Neural Network

- NN is a processing device, either an algorithm or a hardware, whose design was inspired by the design and functioning of human brain
- *A neural network is a computational model used to recognize patterns and make predictions or decisions based on data.*
- NN learn by example, which makes it more flexible and powerful
- There is no need to devise an algorithm to perform specific task
- There is no need to understand the internal mechanism of the task
- They are well suited for real world problems

Artificial Neural network

- An artificial neural network or ANN uses layers of interconnected nodes called neurons that work together to process and learn from the input data.

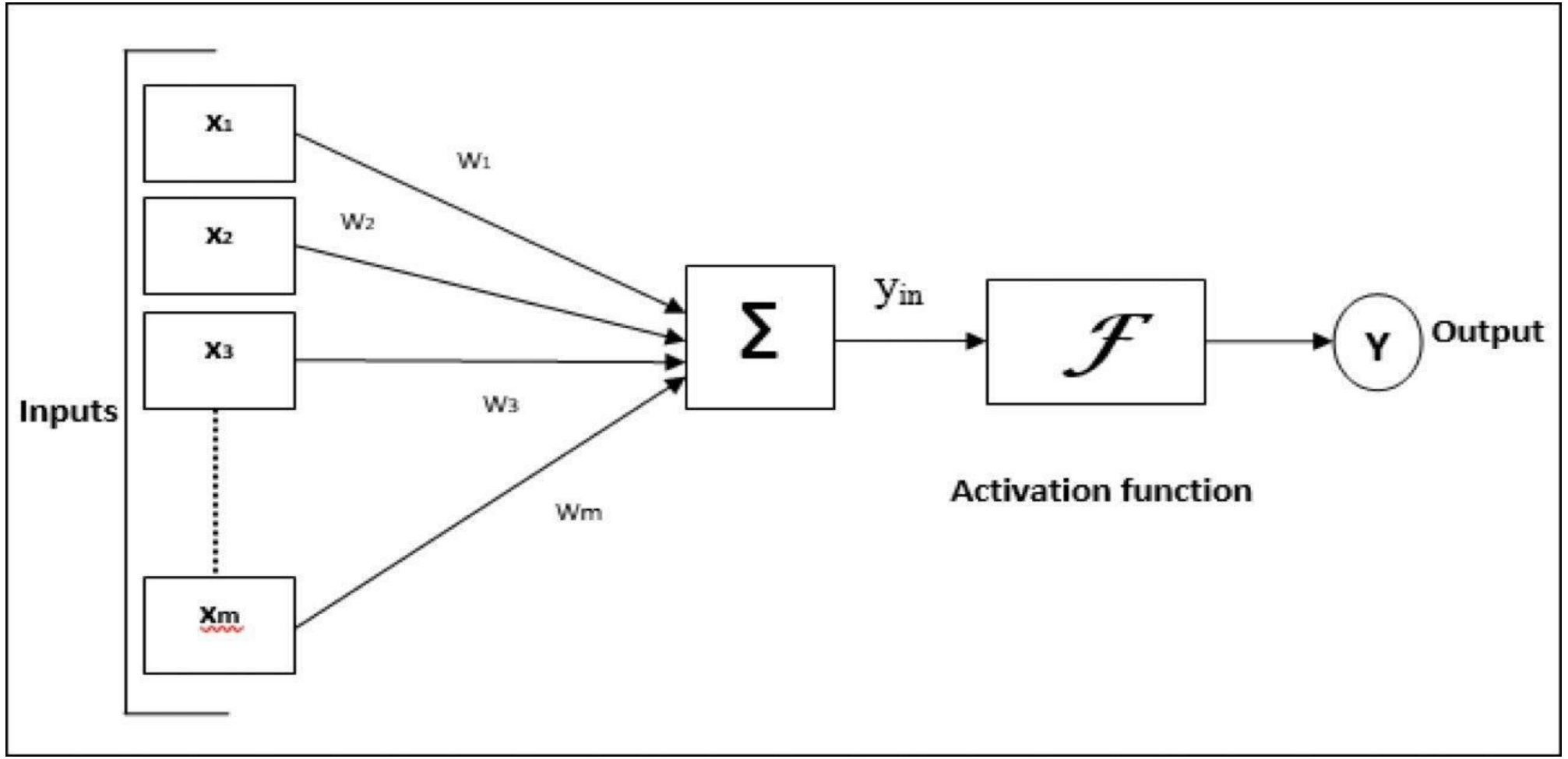




- Artificial Neural Network has an input layer, an output layer as well as hidden layer.
- first layer, receives input from external sources and passes it onto the hidden layer.
- Each neuron in the hidden layer gets information from the neurons in the previous layer, computes the weighted total, and then transfers it to the neurons in the next layer –to the output layer.
- The connections between layers are weighted.

- In Deep neural network, there is an input layer and one or more hidden layers connected one after the other.
- The whole Artificial Neural Network is composed of artificial neurons, which are arranged in a series of layers.
- Neurons are linked to one another from one layer to another in the bulk of neural networks

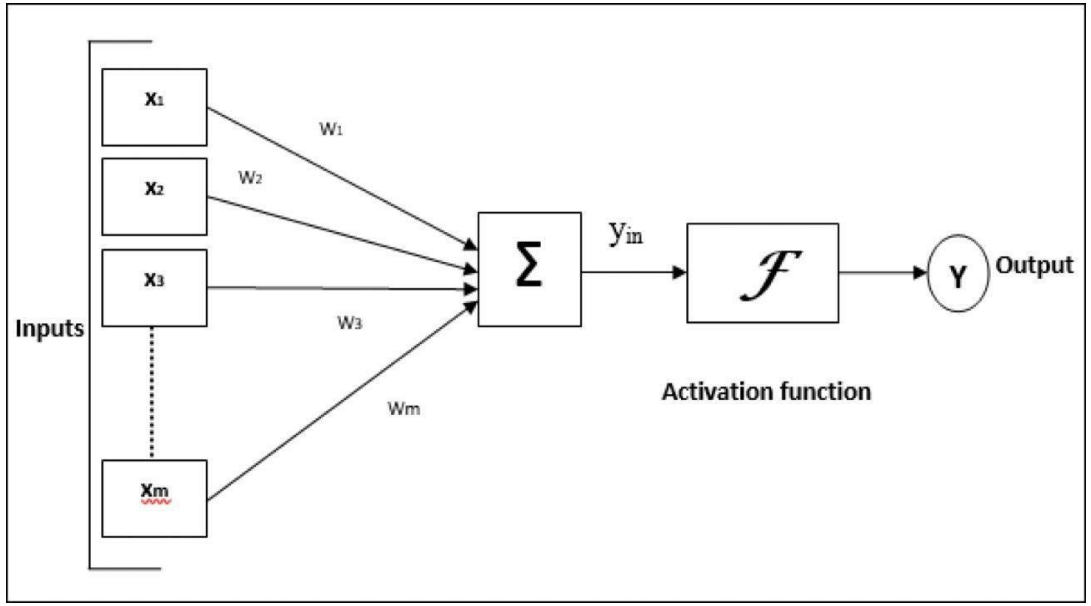
Modern Artificial Neural Network



Elements of a Neural Network

- Input Layer :- This layer accepts input features. It provides information from the outside world to the network, no computation is performed at this layer, nodes here just pass on the information(features) to the hidden layer.
- Hidden Layer :- Nodes of this layer are not exposed to the outer world, they are the part of the abstraction provided by any neural network. Hidden layer performs all sort of computation on the features entered through the input layer and transfer the result to the output layer.
- Output Layer :- This layer bring up the information learned by the network to the outer world.

- The net input can be calculated as follows −
- yin=x1.w1+x2.w2+x3.w3…xm.wm
- The output can be calculated by applying the activation function over the net input.
- Y=F(yin)
- Output = function (net input calculated)



How does Neural Network learn?

- The NN learns by two different ways:
  - Forward Propagation - making a prediction
  - Backpropagation - learning from the error

Forward Propagation

- The data flows through the network beginning at the input layer, progressing through the hidden layers, and ending at the output layer — a process known as forward propagation.
- At each layer, neurons carry out mathematical computations, typically applying an activation function to introduce non-linearity.
- This non-linearity enables the network to learn and represent complex patterns beyond simple linear relationships.

Backpropagation

- After the output is generated, the network compares it to the correct output (the **target**) and calculates the **error**.
- **Backpropagation** is the process of sending the error back through the network to adjust the weights of the connections between neurons. The goal is to reduce this error over time, which is done using optimization algorithms like **gradient descent**.
- This process is repeated many times during training, with the weights being adjusted slightly each time, until the network is able to make predictions that are accurate enough for the task.

Building blocks

Processing of ANN depends upon the following:

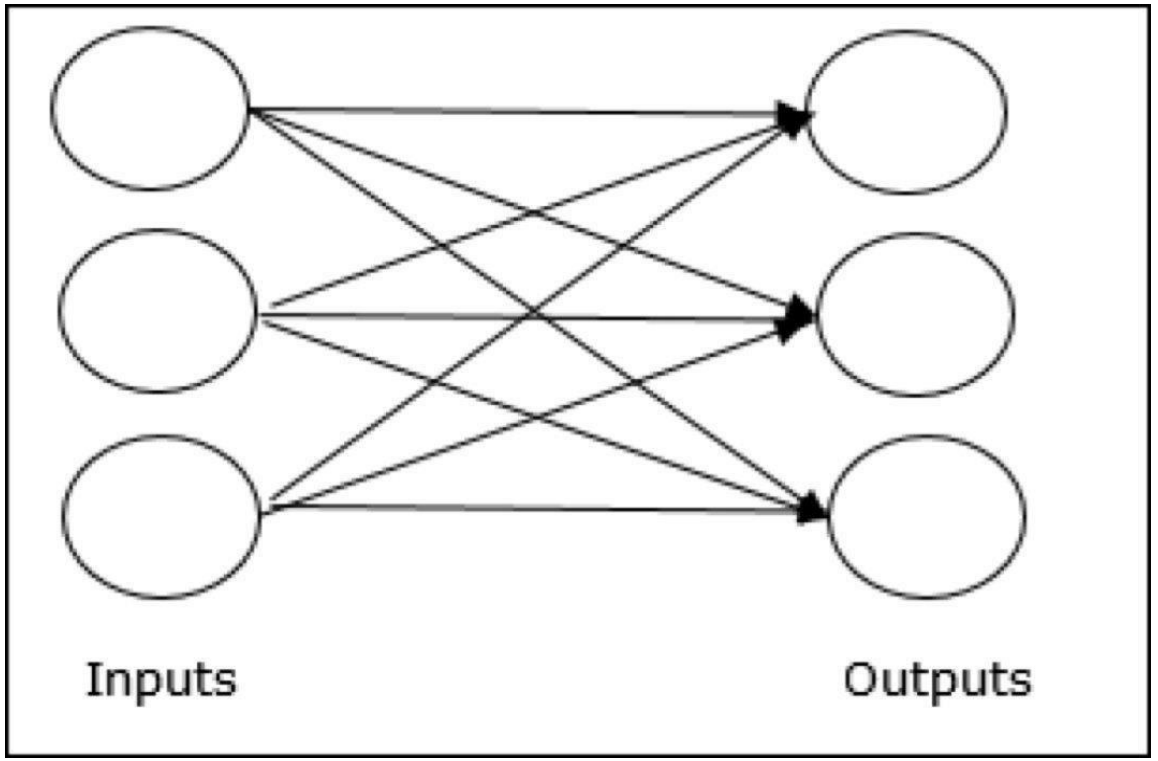
- Network Topology
- Adjustments of Weights or Learning
- Activation Functions

Network Topology

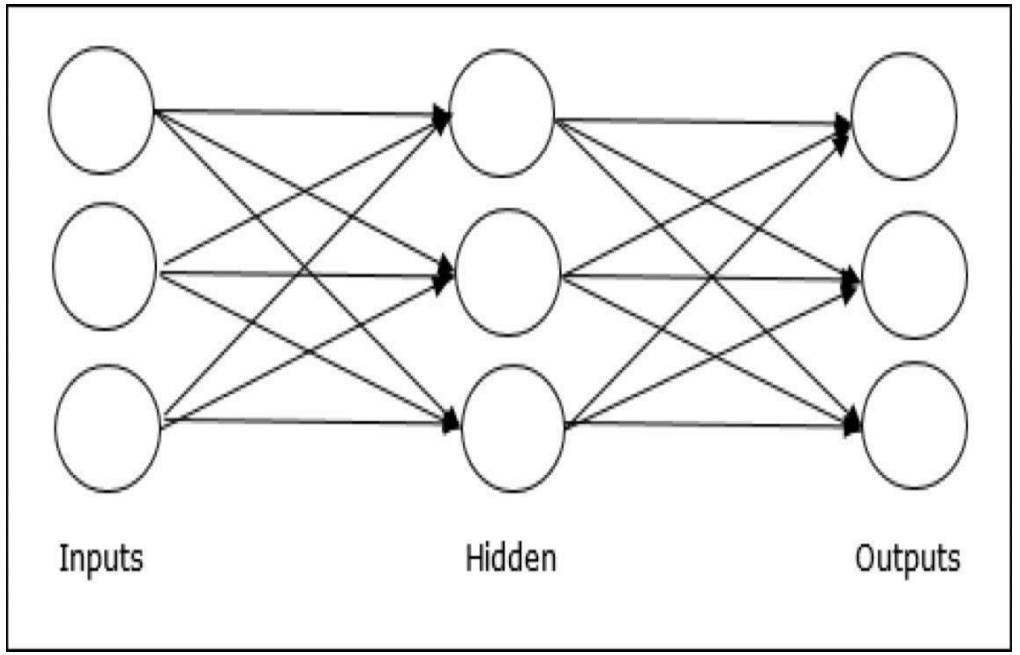
- A network topology is the arrangement of a network along with its nodes and connecting lines. According to the topology, ANN can be classified as the following kinds −
- Feedforward Network
- Feedback Network

Feedforward Network

- **Single layer feedforward network** − The concept is of feedforward ANN having only one weighted layer. In other words, we can say the input layer is fully connected to the output layer.

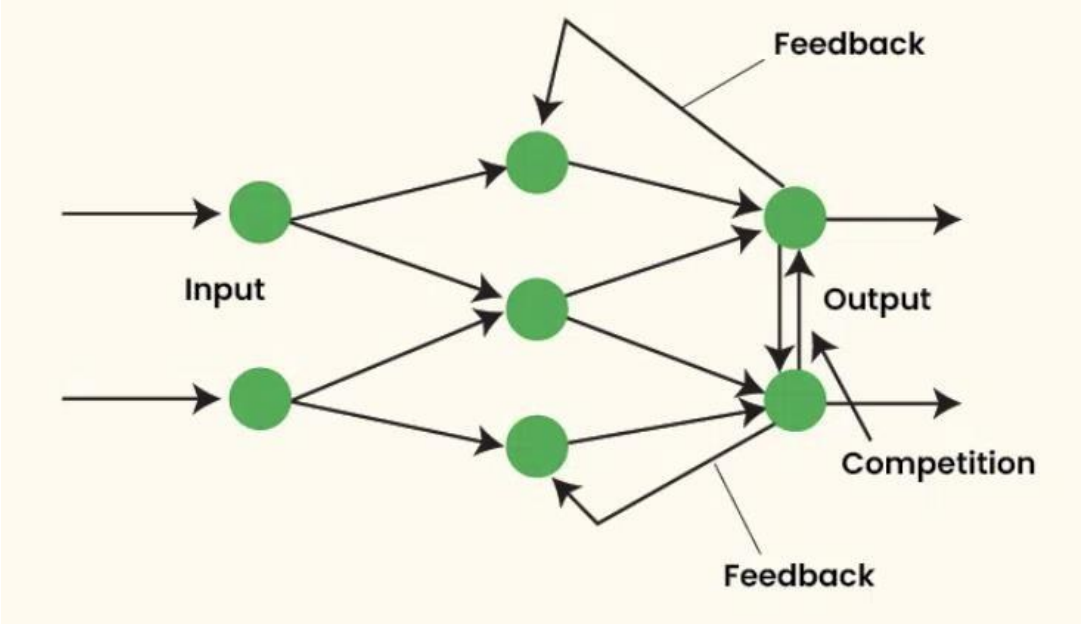


- **Multilayer feedforward network** − The concept is of feedforward ANN having more than one weighted layer. As this network has one or more layers between the input and the output layer, it is called hidden layers.



Feedback Network

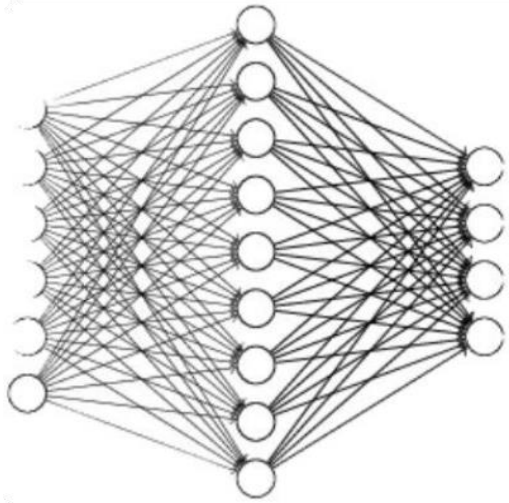
- As the name suggests, a feedback network has feedback paths, which means the signal can flow in both directions using loops. This makes it a non-linear dynamic system, which changes continuously until it reaches a state of equilibrium. It may be divided into the following types −
- **Recurrent networks** − They are feedback networks with closed loops. Following are the two types of recurrent networks.
- **Fully recurrent network** − It is the simplest neural network architecture because all nodes are connected to all other nodes and each node works as both input and output.

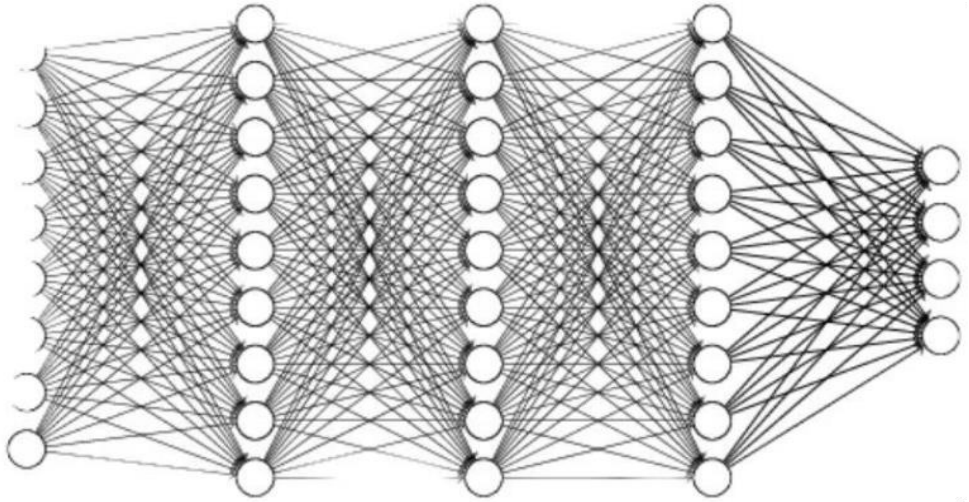


Deeper or wider

The width and depth of Neural Network

- **Depth:** the number of layers in the neural network
- **Width:** the number of neurons in a layer





Wider NN

- A wider network, can potentially offer greater expressive power at each layer and the ability to capture diverse features simultaneously.
- A wider network can process more features in parallel, which can accelerate training and inference times, making it beneficial for applications requiring near real-time processing.
- Wider networks can enhance their capacity to generalize across different data distributions and potentially make them more robust to adversarial attacks

Deeper NN

- Depth emphasizes the sequential processing of information.
- Deeper networks with more layers tend to perform better on more complex tasks, but they also demand more training data and longer training durations.
- But the challenges are Vanishing Gradient, Overfitting and Computational demands

Machine Learning	Deep Learning
Apply statistical algorithms to learn the hidden patterns and relationships in the dataset.	Uses artificial neural network architecture to learn the hidden patterns and relationships in the dataset.
work on the smaller amount of dataset	Requires the larger volume of dataset compared to machine learning
Better for the low-label task.	Better for complex task like image processing, natural language processing, etc.
A model is created by relevant features which are manually extracted from images to detect an object in the image.	Relevant features are automatically extracted from images
It can work on the CPU or requires less computing power as compared to deep learning	It requires a high-performance computer with GPU.

Multilayer Perceptron

- Also known as Deep Feedforward Neural Networks/ Feedforward Neural Networks.
- A multilayer perceptron(MLP) is a type of feedforward neural network consisting of fully connected neurons with a nonlinear kind of activation function.
- It is widely used to distinguish data that is not linearly separable.

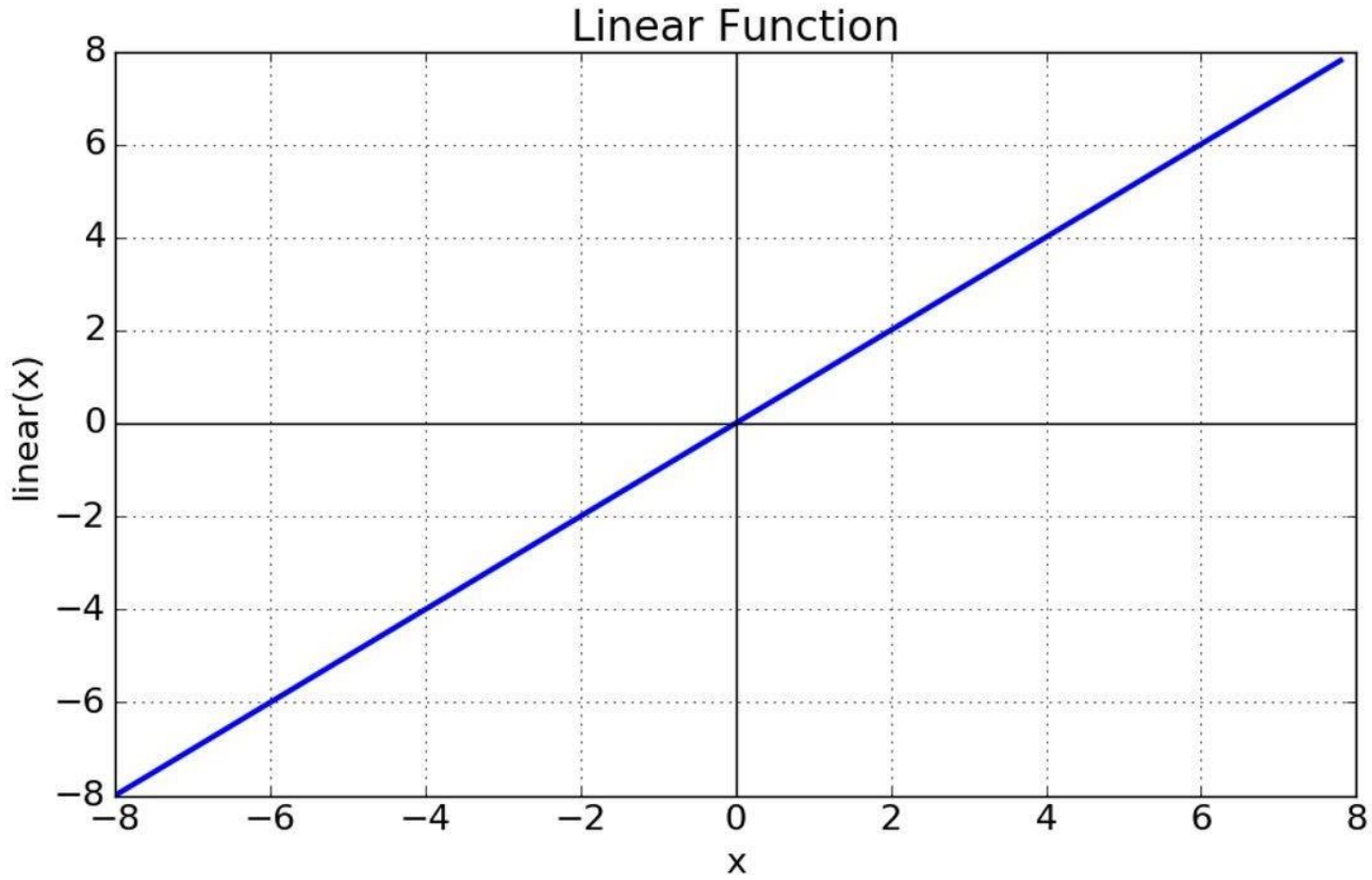
Activation Function

- It is used to determine the output of neural network like yes or no.
- It maps the resulting values in between 0 to 1 or -1 to 1 etc. (depending upon the function).
- The Activation Functions can be basically divided into 2 types-

1.Linear Activation Function

1.Non-linear Activation Functions

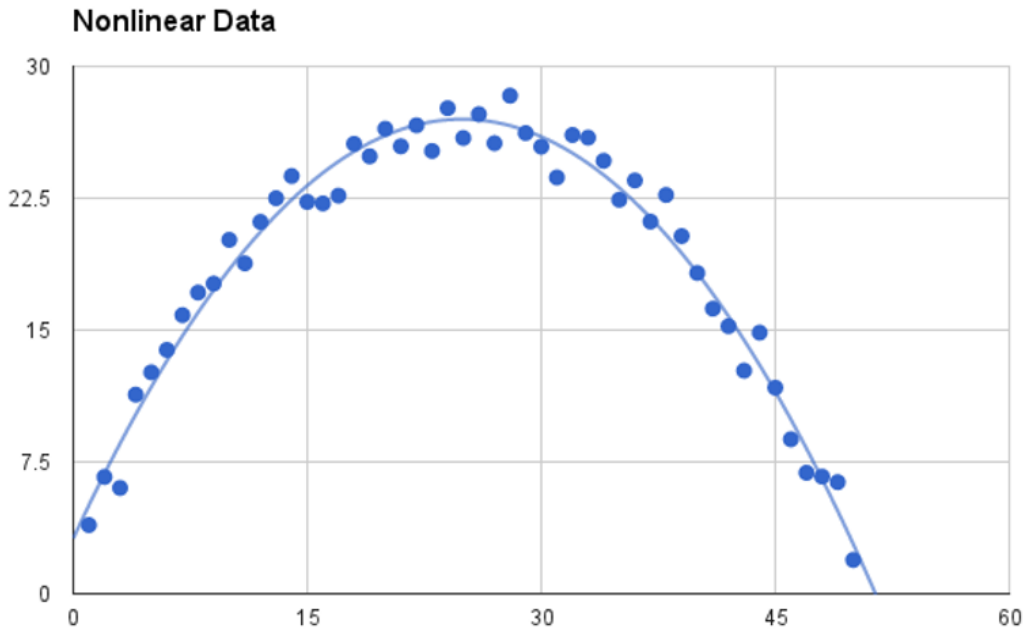
Linear or Identity Activation Function



Equation : f(x) = x

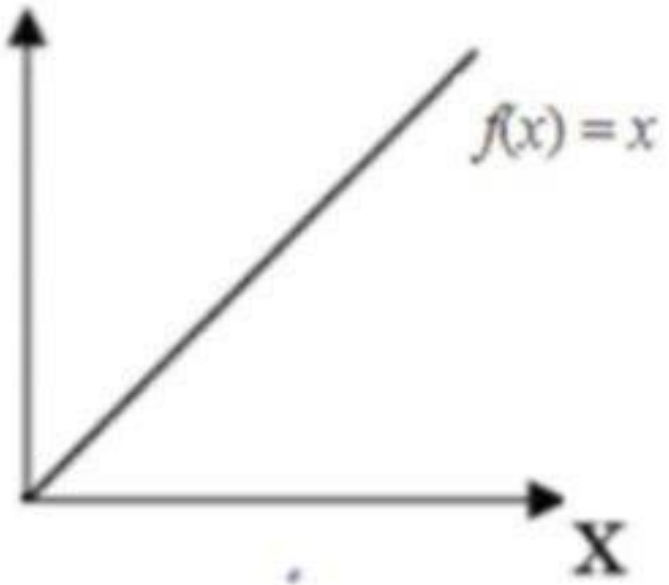
Non-linear Activation Function

- The main terminologies needed to understand for nonlinear functions are:
- ***Derivative or Differential:*** *Change in y-axis w.r.t. change in x-axis. It is also known as slope.*
- ***Monotonic function:*** *A function which is either entirely non-increasing or non-decreasing.*

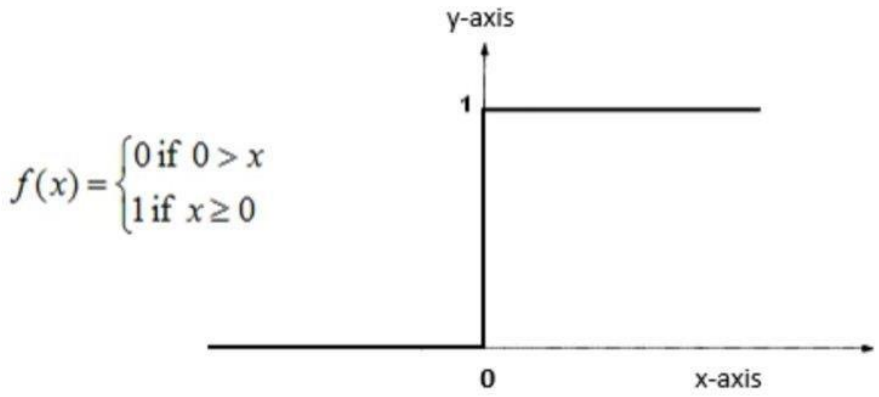


Linear Activation Function

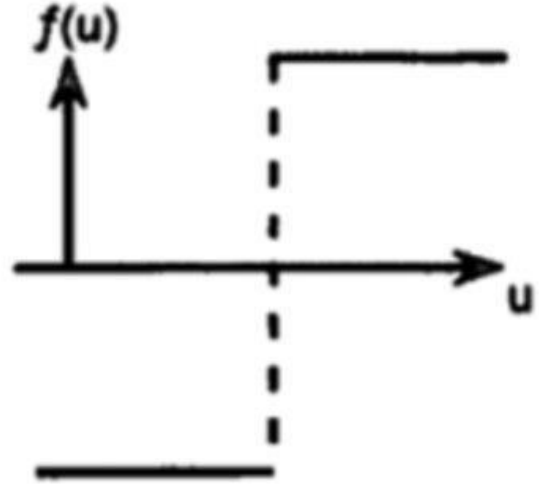
- ***Identity Function***
- Identity function is used as an activation function for the input layer.
- It is a linear function having the form: f(x) = x, for all x
- The output here remains same as input



- **Binary step function**
- if the value of Y is above a certain value known as the threshold, theoutput is True(or activated) and if it's less than the threshold then theoutput is false (or not activated).



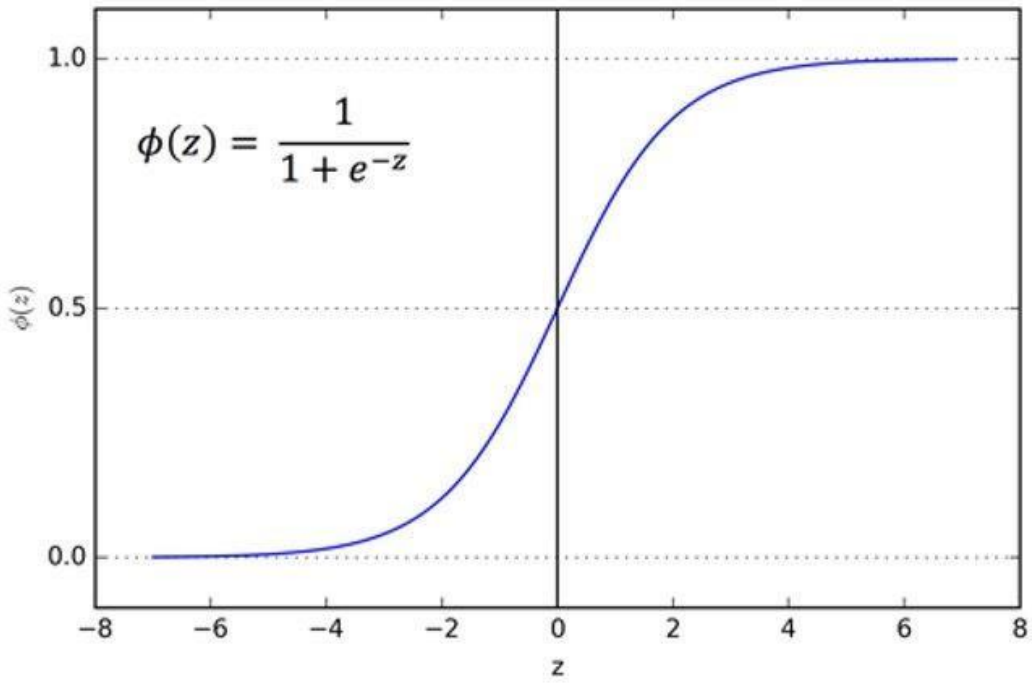
- **Bipolar Activation Function**
- if the value of Y is above a certain value known as the threshold, the output is +1 and if it's less than the threshold then the output is -1. It has bipolar outputs (+1 to -1). It can be utilized in single-layer networks.



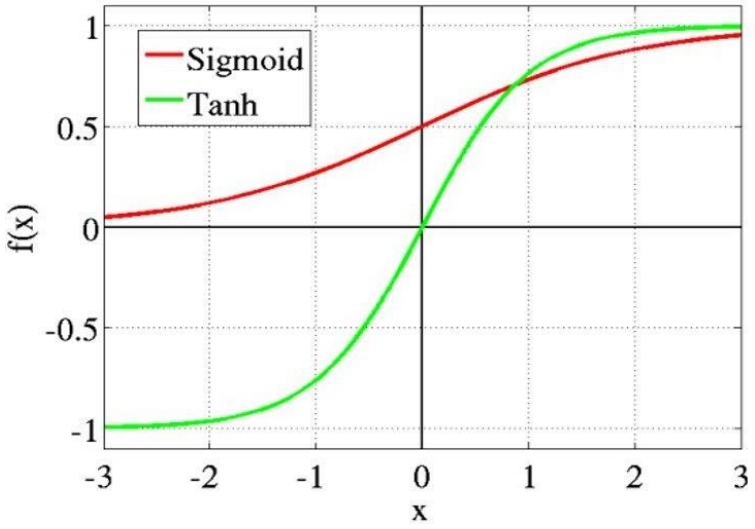
$$f(x) = \begin{cases} 1 \text{ if } x >= t \\ -1 \text{ if } x < t \end{cases}$$

1. Sigmoid or Logistic Activation Function

- The Sigmoid Function curve looks like a S-shape.



- The main reason why we use sigmoid function is because it exists between **(0 to 1).**
- Therefore, it is especially used for models where we have to **predict the probability** as an output.
- Since probability of anything exists only between the range of **0 and 1,** sigmoid is the right choice.



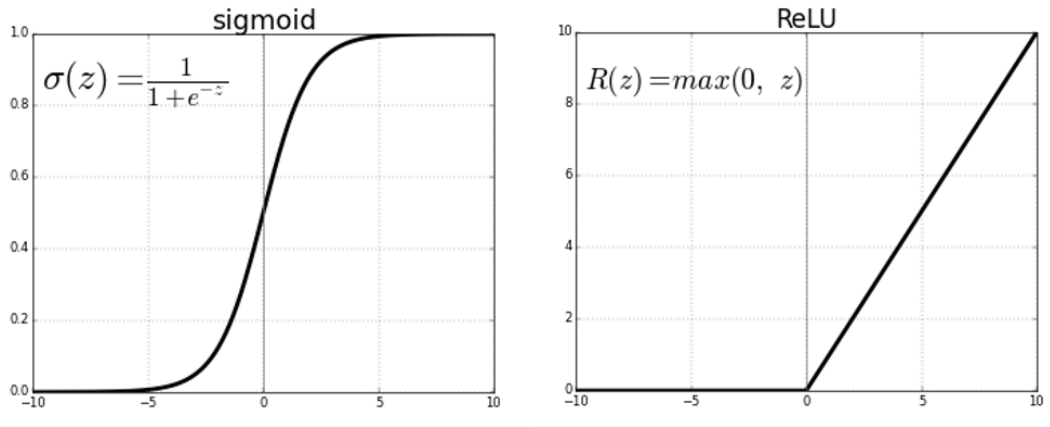
Tanh or hyperbolic tangent Activation Function

- tanh is also like logistic sigmoid but better.
- The range of the tanh function is from (-1 to 1). tanh is also sigmoidal (s - shaped).
- The advantage is that the negative inputs will be mapped strongly negative and the zero inputs will be mapped near zero in the tanh graph.

$$f(z) = \frac{e^{z} - e^{-z}}{e^{z} + e^{-z}}$$

ReLU (Rectified Linear Unit) Activation Function

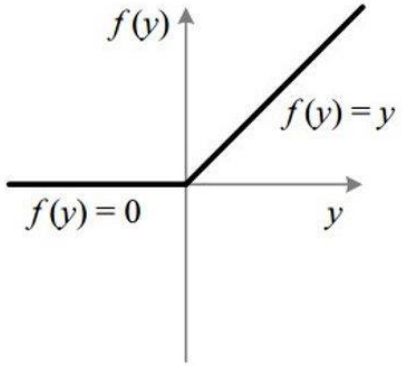
- The ReLU is the most used activation function in the world right now. Since, it is used in almost all the convolutional neural networks or deep learning.
- As you can see, the ReLU is half rectified (from bottom). f(z) is zero when z is less than zero and f(z) is equal to z when z is above or equal to zero.

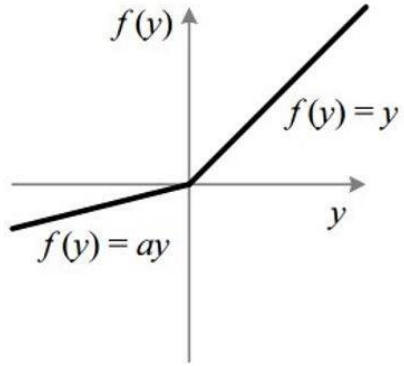


- But the issue is that all the negative values become zero immediately which decreases the ability of the model to fit or train from the data properly.
- That means any negative input given to the ReLU activation function turns the value into zero immediately in the graph, which in turns affects the resulting graph by not mapping the negative values appropriately.

Leaky ReLU

- It is an attempt to solve the dying ReLU problem





- The leak helps to increase the range of the ReLU function. Usually, the value of **a** is 0.01 or so.
- When **a is not 0.01** then it is called **Randomized ReLU**